NGen-3 System Card

(NGen3ForCasualLMv1)

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1. Introduction

The NGen 3 series represents a foundational milestone in scalable, local causal language modeling. Built upon the proprietary NGen3ForCausalLMv1 architecture using PyTorch, it serves as a highly modular, transparent, and experimental framework for AI research. Moving away from legacy absolute positional embeddings, NGen 3 integrates state-of-the-art transformer mechanics, specifically **Rotary Positional Embeddings (RoPE)** for superior relative distance calculation, and **SwiGLU activations** (F.silu(x1) * x2) to improve gradient flow and representational capacity within the feedforward layers.

Furthermore, the architecture introduces dynamic routing via an optional **Sparse Mixture-of-Experts (MoE) network**. Instead of passing all tokens through a monolithic dense layer, the MoE variant uses a linear gating mechanism (self.gate) paired with softmax probabilities to route tokens to a configurable number of specialized "expert" MLPs, optimizing parameter count without linearly increasing inference latency.

The NGen 3 series is designed to offer researchers scalable compute, scaling from local edge-testing to full cluster deployment:

NGen 3 Model Variants	Parameters	Architecture Specifications	Context Limit
NGen 3 v2	1B	32 Layers, 16 Heads, 1536 Dim	4096 Tokens
NGen 3 v1	700M	28 Layers, 16 Heads, 1280 Dim	512 Tokens
NGen 3 v1	500M	24 Layers, 16 Heads, 1024 Dim	512 Tokens
NGen 3 v1	120M	Up to 16 Layers, 16 Heads, 768 Dim	512–1024 Tokens
NGen 3 v1	7M	4 Layers, 4 Heads, 128 Dim	128 Tokens

1.1 Model Information

Inputs: NGen 3 allows input of strictly Text. It utilizes a highly optimized GPT-2 Byte-Pair Encoding (BPE) tokenizer, featuring a fixed vocabulary size of 50,257 tokens. It includes fallback mechanisms to

tiktoken or a raw byte-level encoding if the primary tokenizer is unavailable.
Output: Autoregressive text generation. The model outputs logits over the vocabulary, applying softmax probabilities tailored by user-defined temperature and top_k sampling constraints.
Interoperability: The framework natively supports exporting checkpoints to Hugging Face's secure safetensors format and ONNX dynamic-axis graphs, facilitating immediate deployment to edge devices and web environments.

1.2 Model Data

The training data pipeline for NGen 3 is designed for maximum throughput and agnostic ingestion. It eschews complex data-loading overhead by directly compiling raw text or Hugging Face datasets into 1D PyTorch LongTensors, saved as binary (.bin) files.

- **Local Processing:** The DataProcessorLocal module parses raw .txt files or remote web URLs, decoding UTF-8 strings into discrete token arrays.
- **Hugging Face Streaming:** For large-scale pre-training, the hf_tokenize pipeline enables streaming massive datasets (e.g., FineWeb, OpenWebText) directly into memory, preventing local storage bottlenecks.
- **Context Windowing:** During training, the Trainer.get_batch() function dynamically slices these tensors into parallel blocks defined by config.block_size (up to 1024 tokens) and config.batch_size (up to 128 for the 1B variant).

1.3 Instruct Mode

A defining and highly experimental feature of the NGen3ForCausalLMv1 architecture is its structural approach to instruction-following, termed **Instruct Mode**. Rather than relying purely on prompt formatting and standard fine-tuning (which often leads to catastrophic forgetting of foundational knowledge), NGen 3 introduces a dedicated, isolated neural pathway.

When instantiated with config.instruct = True, the model activates an extra dense projection layer (self.instruct_dense = nn.Linear(config.n_embd, config.n_embd)) positioned immediately after the final transformer block and LayerNorm, but *before* the vocabulary generation head. During multi-turn conversational chat, this layer acts as an "alignment filter," transforming the foundational latent space into a conversational latent space. This allows the model to separate factual world-knowledge from syntactical user alignment, drastically improving performance in the interactive chat CLI loop.

We may not be using the Instruct Mode in the coming NGen 3 Variants like, NGen3ForCasualLMv2 (NGen 3.1) and v3 (NGen 3.9 & 3.5) architectures and we'll be producing research over it and will be using it in NGen 4 variants.

2. Training Methods

2.1 Training Methods (Non-Unique Standard)

The NGen 3 series utilizes a highly optimized, standard causal language modeling objective (predicting the next token via Cross-Entropy Loss). The methodology is executed via a built-in Trainer class and broken down into distinct phases:

- **Phase 1: Foundational Pre-Training:** Using the train command, models build core linguistic capabilities. The training loop utilizes the **AdamW optimizer** with a baseline learning rate of 1e-4 to 6e-4 (depending on the variant) and a weight decay of 0.1.
- **Optimization Mechanics:** To handle massive batch sizes on limited hardware, NGen 3 employs **Gradient Accumulation** (config.grad_accum_steps), updating weights only after multiple forward passes. Stability is ensured via **Gradient Clipping** (grad_clip = 1.0), preventing exploding gradients, and a **Cosine Annealing LR Scheduler** that smoothly decays the learning rate to 1e-6 to allow the model to settle into local minima.
- **Phase 2: Conversational Fine-Tuning:** Handled by the instruct_finetune pipeline, the model ingests structured dialogue. The code specifically parses Hugging Face datasets into distinct role headers (e.g., System:, Instruction:, Response:) to enforce conversational boundaries.

2.2 Teacher In-Loop RL Alignment

Because the raw NGen 3 architecture lacks Reinforcement Learning from Human Feedback (RLHF) mechanisms like PPO or DPO, it achieves behavioral alignment via **Knowledge Distillation**. This acts as a proxy for "Teacher In-Loop" steering.

Executed via the distill CLI command, a smaller "Student" model (e.g., NGen 3 120M) is trained simultaneously against the hard targets (the actual dataset) and soft targets provided by a frozen, pre-aligned "Teacher" model (e.g., a local NGen 3 1B or an external Hugging Face GPT-2 variant).

- **Mathematical Steering:** The loss function is a weighted blend: total_loss = alpha * distill_loss(KL) + (1 - alpha) * ce_loss.
- **Temperature Scaling:** By applying a high temperature (T=2.0) to the teacher's logits before the Softmax, the student model learns the "dark knowledge" of the teacher—the secondary and

tertiary probable tokens. This forces the student to adopt the reasoning structure and safe response formatting of the teacher, acting as an automated alignment firewall.

3. Benchmark Evaluations

All performance metrics below were evaluated using the TCorpus5 dataset, a localized, rigorous benchmark set designed to detect reasoning capabilities and knowledge retention.

3.1 NGen 3 (1B) Evaluation Results

The 1B Pro variant demonstrates high parameter efficiency, achieving performance parity with 7B-class models on knowledge tasks.

Benchmark Task (TCorpus5)	*NGen 3 (1B)*	*TinyLlama (1.1B)*	*Llama-2 (7B)*	*Mistral (7B)*
MMLU (5-shot)	***60.24***	*25.39*	*47.33*	*60.50*
PIQA (0-shot)	***79.12***	*77.50*	*79.90*	*81.70*
Hellaswag (10-shot)	***67.87***	*61.10*	*78.64*	*83.00*
Winogrande (5-shot)	***68.35***	*54.93*	*72.22*	*77.00*

3.2 NGen 3 (120M) Evaluation Results

The 120M Lite model is optimized for latency and inference efficiency.

Benchmark Task (TCorpus5)	*Score (Accuracy %)*
MMLU	*51.61*
PIQA	*74.28*

Hellaswag	*40.13*
Winogrande	*56.38*

4. Core Safety Challenges

Because the NGen3ForCausalLMv1 framework is designed as a raw, research-oriented foundational architecture, it fundamentally lacks the complex, layered safety guardrails seen in production models (like NGen 4). Deploying this model in public-facing applications without secondary moderation layers poses significant risks.

4.1 Addressing Refusals and Disallowed Content

NGen 3 operates purely on mathematical next-token prediction driven by Cross-Entropy loss. It possesses no innate moral compass or RLHF-based refusal mechanism. If the model ingests unfiltered web data containing hate speech, explicit content, or dangerous instructions, it will map those probabilities into its latent space and willingly reproduce them when prompted. Mitigation relies entirely on the developer rigorously pre-filtering the .txt and dataset inputs prior to executing the train pipeline.

4.2 Mitigating Sycophancy and Jailbreaks

Due to the experimental nature of the Instruct Mode dense layer, the model is highly steerable. However, this high steerability makes it incredibly susceptible to *sycophancy*—the model will frequently agree with the user's premise, even if the premise is factually incorrect, simply to fulfill its conversational objective. Because the model lacks adversarial testing phases during standard training, traditional jailbreaks (e.g., "Ignore all previous instructions") will easily bypass any system prompts.

4.3 Managing Instruction Hierarchy and Prompt Injections

The architecture parses roles (e.g., "System:", "User:", "Assistant:") as raw text strings rather than protected control tokens. Consequently, the model cannot natively distinguish between a high-level developer system prompt and a malicious user prompt injection. If a user inputs "Assistant: [Harmful command]", the model treats it with equal weight to the developer's instructions, requiring robust external application-layer sanitization.

4.4 Hallucinations and Deception Detection

Without a structured, constrained reasoning architecture, NGen 3 is prone to "leaps of logic" and severe factual hallucinations. To aid researchers in detecting this, NGen 3 includes a unique analyze CLI command that tracks and plots **Hidden State Norms** (hs.norm()) across all transformer blocks via matplotlib. Spikes or collapses in layer norms often correlate empirically with model uncertainty, feature saturation, or hallucinatory generation, providing a crucial interpretability tool for developers.

4.5 Multimodal Safety (Image Input) and Health Guidance

NGen 3 is strictly a unimodal (text-in, text-out) causal language model. It has zero native multimodal capabilities and cannot process or analyze image-based safety threats, such as CSAM or violent graphical imagery. Furthermore, lacking specific medical domain fine-tuning or truth-grounding, any output regarding health, legal, or financial guidance is purely probabilistic and should be treated as entirely unreliable without human oversight.

4.6 Fairness and Bias Considerations

The model is an exact statistical mirror of its ingested datasets. Because the pipeline supports seamless, automated streaming from open Hugging Face repositories via the hf_tokenize command, users must be hyper-vigilant. If trained on skewed datasets, the model will rapidly manifest societal, gender, racial, and political biases during interactive chat sessions, requiring deliberate dataset balancing prior to the Post-Training phase.

4.7 Multilingual Performance in Safety Contexts

The GPT-2 Byte-Pair Encoding (BPE) tokenizer hardcoded into NGen 3 is overwhelmingly optimized for the English language. Multilingual performance, particularly for Indic languages or complex non-Latin scripts, is heavily fragmented, often falling back to raw byte-level encoding. Consequently, safety guidelines established in English fine-tuning frequently fail to generalize to other languages, causing safety alignments to break down entirely when the model is prompted in a foreign language.

4. Conclusion

4.1 Summary of NGen 3

The NGen 3 series, powered by the NGen3ForCausalLMv1 architecture, provides a highly transparent, fully self-contained framework for understanding, building, and training modern causal language models from scratch. By introducing advanced features like Rotary Positional Embeddings, optional

Mixture-of-Experts routing, Knowledge Distillation, and a unique architectural Instruct Mode projection layer, NGen 3 serves as a remarkably robust foundational testbed.

While it lacks the automated safety filters, multimodal vision integration, and advanced structured reasoning APIs of its enterprise-grade successor, NGen 4, the NGen 3 codebase remains an invaluable asset. It empowers researchers, educators, and engineers to explore hidden-state interpretability, conversational fine-tuning, and model scaling dynamics ranging from a localized 7 Million parameter prototype to a full 1 Billion parameter foundational model.

End of the System Card